

WORKING PAPER

An Auditable Public-Data U.S. Equity Research Pipeline: Point-in-Time Factors, Asset Pricing, Portfolio Construction, and Walk-Forward Machine Learning

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Abstract

This paper studies a public-data U.S.-equity research pipeline in which security identity, filing availability, source transitions, transaction costs, and temporal validation are specified before performance is interpreted. The released sample screens 822 securities and admits 487 equities. It contains 715,447 daily market observations and 37,273 point-in-time fundamental observations. Forty-seven of 48 factor definitions are estimable. Five asset-pricing families show higher mean in-sample adjusted R-squared for multifactor specifications (0.259–0.301) than for CAPM (0.063). Eight long-only portfolio methods are evaluated under ex-ante one-way costs of 5, 10, and 20 basis points over a 32-month factor-portfolio window. Eight prediction models are assessed in 17 purged monthly test folds using two characteristics. Every non-constant model's bootstrap interval for mean rank correlation includes zero, and stable after-cost incremental value is not established. The contribution is a connected empirical design that preserves negative and inconclusive findings while making the sample's public-data and short-window limitations explicit.

Keywords: empirical asset pricing; point-in-time data; factor models; portfolio construction; transaction costs; walk-forward validation; machine learning; reproducibility

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1. Introduction

1.1 Motivation

Empirical finance often begins after the hardest data decisions have already been made. Yet security identity, delisting coverage, accounting availability, corporate-action treatment, and source changes determine which observations a model sees. Portfolio and machine-learning layers add further choices about costs, sample splits, tuning, and interpretation. This study places those decisions inside one versioned workflow and asks what conclusions remain after the evidence is bounded explicitly.

1.2 Research problem

The project uses free and public sources rather than CRSP or Compustat. That choice expands reproducibility and access, but it also creates limitations that cannot be repaired by software alone. A fixed upstream ticker universe may omit firms that disappeared before the provider snapshot; a change from consolidated-tape to IEX-only bars alters volume comparability; and SEC facts become usable only after filing acceptance. The empirical problem is therefore both statistical and evidentiary.

1.3 Research questions

The analysis asks whether factor construction is feasible on the selected panel, whether multifactor models improve explanatory fit, how long-only allocations differ under fixed costs, whether costs reduce observed cumulative outcomes, and whether two-characteristic machine-learning models add stable out-of-sample information or economic value.

1.4 Contributions

The paper contributes a connected public-data study rather than a new factor theorem. It combines effective-dated identity, point-in-time accounting evidence, a broad factor registry, robust asset-pricing inference, eight constrained allocation methods, and purged walk-forward prediction. It also treats non-estimability, weak prediction, and incomplete hypothesis decisions as legitimate outcomes.

1.5 Paper structure

Sections 2–5 develop the literature, data, sample, and design. Sections 6–9 present factor, asset-pricing, portfolio, and machine-learning evidence. Sections 10–15 evaluate robustness, hypotheses,

implications, limitations, future work, and conclusions. Technical identities and extended tables appear in Appendices A–M.

2. Literature Review

2.1 Asset pricing and factor models

Markowitz (1952) formalized portfolio choice under mean and variance, and Sharpe (1964) connected systematic risk to expected return in the CAPM. Fama and French (1993) expanded the empirical model with size and value factors; Carhart (1997) added momentum; Fama and French (2015) introduced profitability and investment factors; and Hou, Xue, and Zhang (2015) developed a q-factor interpretation. These models motivate the five pre-specified families evaluated here, but higher explanatory fit is not treated as proof of persistent investable alpha.

2.2 Portfolio construction and estimation error

Mean-variance optimization is sensitive to expected-return and covariance estimation. DeMiguel, Garlappi, and Uppal (2009) show why simple diversification can remain difficult to beat out of sample. Maillard, Roncalli, and Teiletche (2010) develop equal-risk-contribution portfolios, López de Prado (2016) proposes hierarchical risk parity, and Rockafellar and Uryasev (2000) provide a tractable conditional-value-at-risk formulation. The present comparison therefore spans naive, moment-based, diversification, risk-budgeting, hierarchical, and tail-risk approaches.

2.3 Statistical inference and multiple testing

White (1980) and Newey and West (1987) motivate covariance estimators robust to heteroskedasticity and, for HAC inference, serial dependence. Harvey, Liu, and Zhu (2016) emphasize that a large factor search raises false-discovery risk. Because the released Phase-2 publication lacks the complete factor-level statistics needed for the original multiplicity rule, the corresponding hypothesis is left inconclusive rather than reconstructed after the fact.

2.4 Financial machine learning and temporal validation

Ridge regression (Hoerl and Kennard, 1970), Elastic Net (Zou and Hastie, 2005), Random Forest (Breiman, 2001), and XGBoost (Chen and Guestrin, 2016) represent regularized linear and nonlinear prediction families. Gu, Kelly, and Xiu (2020) document the potential of machine learning in asset pricing, while Bailey et al. (2014) and López de Prado (2018) stress temporal validation and backtest-overfitting controls. SHAP supplies fitted-model attributions (Lundberg and Lee, 2017), not causal effects.

3. Data Sources and Source Limitations

Table 1. Data sources and principal limitations

Source	Role	Temporal use	Principal limitation
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Source	Role	Temporal use	Principal limitation
HF Data Library	Market histories and candidate universe	Daily bars aggregated from intraday evidence	Fixed upstream universe before 2022; PiTrading/IEX source transition
SEC EDGAR	Filings, XBRL facts, ticker/CIK evidence	Actual acceptance timestamps	Taxonomy and concept heterogeneity
Alpha Vantage	Listing-status and lifecycle evidence	Historical lifecycle support; coverage may reach 2010	Not the start date of the empirical market panel
FRED DGS3MO	Three-month Treasury-bill reference	Frequency-matched risk-free series	Yield must be mapped to period returns
Kenneth French Data Library	Factor concepts and reference evidence	Daily/monthly where specified	Not a security master
SPY	Broad S&P 500 total-return proxy	Benchmark and 21-session relative ML target	ETF proxy rather than index database

3.1 HF Data Library

HF Data Library supplied the stock candidate set and market histories used locally. The official dataset record is Elkassabgi (2026). The provider’s current documentation distinguishes pre-March-2022 consolidated-tape evidence from post-March-2022 IEX evidence and requests attribution to the original sources where noted.

3.2 SEC EDGAR

SEC submissions, Company Facts, and tagged filing covers provide issuer facts and their availability times. Period end describes the accounting period; filing acceptance determines when a fact could enter research. Accession, taxonomy, concept, unit, and availability are retained.

3.3 Alpha Vantage lifecycle evidence

The Alpha Vantage LISTING_STATUS endpoint supports active and delisted-symbol evidence at historical dates. Some lifecycle evidence extends to 2010, but the released empirical market panel begins on 14 February 2018. These are distinct coverage statements.

3.4 FRED and Kenneth French

The three-month Treasury-bill series DGS3MO from FRED supplies the approved risk-free reference. Kenneth French’s library supplies established factor conventions and comparison evidence; it does not replace the project’s security-level source data.

3.5 Benchmark convention

SPY is the broad S&P 500 total-return proxy. Security-specific benchmark gaps are not compounded on different calendars; a common benchmark calendar is used for cross-portfolio comparison.

3.6 HF pre-2022 survivorship limitation

HF Data Library states that its universe was fixed around 2022–2023 and was not reconstructed point in time. Firms that delisted, failed, or were acquired before that snapshot can be absent. The limitation is most severe before 2022 and matters for long-horizon and cross-sectional asset-pricing work. The project’s identity and lifecycle checks can reduce errors among supplied candidates, but they cannot restore securities missing from the upstream universe. The sample is therefore not CRSP-grade survivorship-complete.

3.7 March-2022 PiTrading/IEX structural break

The provider reports a source change in March 2022: earlier bars reflect consolidated-tape PiTrading evidence, whereas later bars derive from IEX. IEX represents approximately 2–3% of consolidated U.S. equity volume. Absolute volume and liquidity measures are therefore not fully comparable across the splice. Average dollar volume, turnover, Amihud illiquidity, and related characteristics require particular caution. Project screening rejects unresolved source splices but cannot eliminate the structural difference in accepted histories.

4. Sample Construction and Point-in-Time Controls

4.1 Candidate universe

All 822 candidates passed through the same screening sequence before factor, portfolio, or prediction results were examined. The frozen sample accepts 487 equities and rejects 335: an acceptance rate of $487/822 = 59.25\%$ and a rejection rate of $335/822 = 40.75\%$. The accepted sample contains 372 Tier A and 115 Tier B securities; Tier C contains none.

Table 2. Pre-specified research hypotheses

Hypothesis	Pre-specified proposition
H1	At least one factor spread survives the complete pre-specified cost and multiplicity rule
H2	Multifactor models improve explanatory fit relative to CAPM
H3	Estimated relations are stable through time under the original rule
H4	Higher transaction-cost assumptions reduce cumulative outcomes
H5	Constrained methods alter risk-return trade-offs
H6	ML adds stable OOS predictive and after-cost value
H7	Explanation rankings are stable across methods

Table 3. Universe-screening outcomes

Disposition	Count	Share of candidates
Accepted	487	59.25%
Rejected	335	40.75%
Tier A	372	45.26%
Tier B	115	13.99%

Disposition	Count	Share of candidates
Tier C	0	0.00%

4.2 Security identity

Listing identity is effective-dated and distinct from issuer identity. Ticker changes, share classes, venue changes, and issuer succession are not collapsed into a timeless present-day ticker.

4.3 Filing availability

Fundamental observations enter only after the recorded SEC availability time. The released fundamental evidence spans 23 February 2019 through 15 August 2026 by `available_at`, while the market panel spans 14 February 2018 through 4 August 2026.

4.4 Shares and split basis

Historical prices and shares must use compatible corporate-action bases. Candidates with unresolved share or split histories are excluded instead of forcing a mechanically complete market-cap series.

4.5 Lifecycle and terminal-event treatment

Terminal outcomes are bounded with available listing and issuer evidence. Unresolved exits remain exclusions. This control addresses supplied candidates only and does not fill the provider's pre-2022 universe omissions.

4.6 Final cross-sectional breadth

Table 4. Annual daily security breadth

Year	Minimum	Median	Mean	Maximum
2018	1	4	3.7	4
2019	4	14	17.8	46
2020	46	333	295.5	357
2021	357	398	393.3	404
2022	404	428	425.2	431
2023	431	463	457.6	466
2024	466	486	483.3	487
2025	487	487	487.0	487
2026	486	487	487.0	487

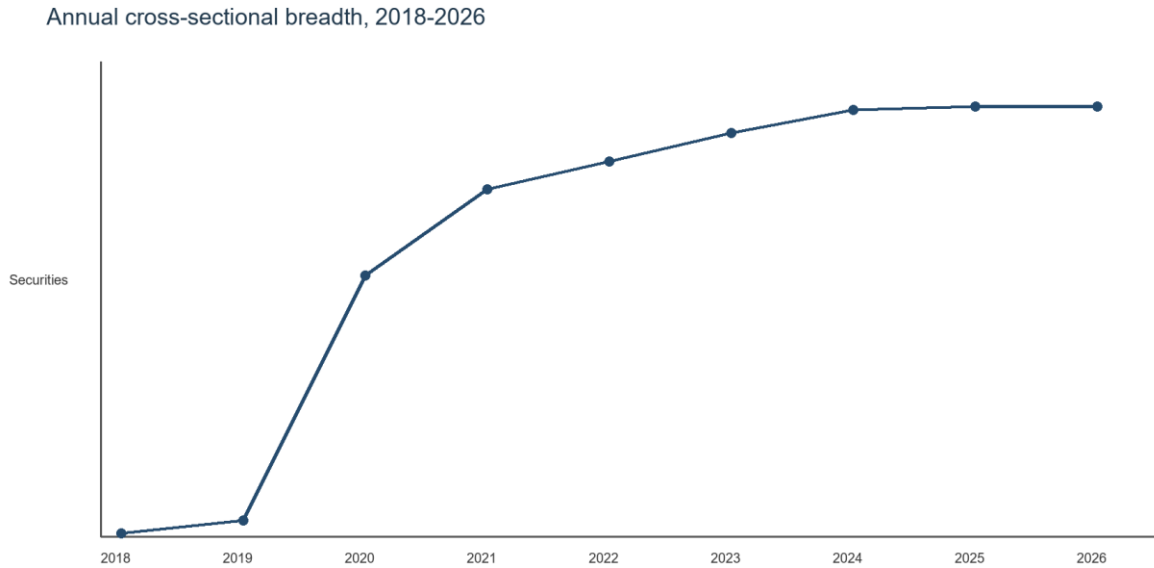


Figure 1. Annual cross-sectional breadth, 2018-2026.

The early panel is intentionally sparse: only evidence available at the date is admitted. Breadth expands sharply in 2020 and reaches 487 by 2024. The 2026 row ends on 4 August. This pattern makes early cross-sectional estimates more fragile than later ones.

4.7 Descriptive statistics

Table 5. Core descriptive statistics from released panels

Variable	N	Mean	SD	P1	P25	Median	P75	P99
security return	715,447	0.0005868	0.02612	-0.07017	-0.01053	0.0005752	0.01157	0.0734
market capitalization	706,285	5.369e+10	1.803e+11	4.923e+08	1.015e+10	2.16e+10	5.017e+10	3.673e+11
average dollar volume	715,447	6.243e+07	3.062e+08	1.533e+05	4.897e+06	1.268e+07	4.232e+07	6.672e+08
book to market	585,078	0.3955	0.4406	-0.2019	0.1188	0.2926	0.5579	1.9823
momentum 12 1m	593,210	0.1250	0.4443	-0.6015	-0.1185	0.0691	0.2833	1.6455
rolling volatility	706,194	0.3899	0.2097	0.1574	0.2521	0.3295	0.4623	1.1676

Market capitalization equals released price multiplied by released shares outstanding; average dollar volume equals price multiplied by volume. The characteristic rows are direct descriptive summaries of the long-form factor IDs book_to_market, momentum_12_1m, and rolling_volatility. They do not represent new factor tests.

5. Research Design and Hypotheses

The workflow is one way: source evidence feeds point-in-time panels, factors, asset-pricing tests, portfolios, and finally walk-forward prediction. Later phases may consume but do not rewrite earlier publications. Universe admission precedes analytical inspection, portfolio costs precede performance reading, and ML configuration precedes final fold evaluation.

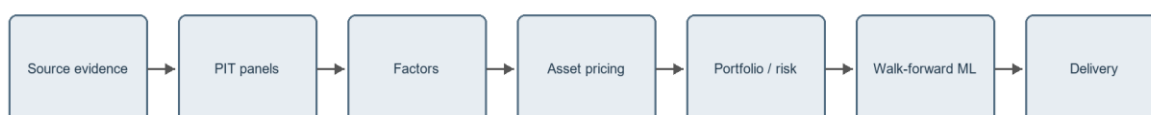


Figure 2. One-way empirical research workflow.

The seven hypotheses permit supported, partially supported, inconclusive, and not-supported outcomes. H1 requires factor-level cost and multiplicity evidence; H3 requires the complete original temporal-stability decision. Missing required outputs are not replaced with post hoc tests.

6. Factor Construction and Results

6.1 Factor families

The registry spans market, size, value, momentum, profitability, investment, accrual, quality, risk, trading, liquidity, growth, payout, leverage, and composite families. Values are winsorized and normalized according to the released configuration, with monthly portfolio formation where specified.

Table 6. Factor families and estimability summary

Evidence	Released result
Configured definitions	48
Estimable definitions	47
Non-estimable definition	equity_issuance
Factor observations	34,341,456
Factor-portfolio observations	11,545
Sample dates	2018-02-14 to 2026-08-04

6.2 Portfolio construction

Cross-sectional ranks and high-minus-low portfolios translate characteristics into comparable evidence. Value-weighted terciles and the common benchmark calendar are implementation choices fixed in the release.

6.3 Missingness and estimability

Coverage differs across characteristics because market histories and accounting inputs become available at different dates. Estimability means a factor can be constructed under the released rule; it does not imply statistical significance.

6.4 Equity issuance limitation

Equity issuance is not estimable because the required issuance and shareholders' equity inputs cannot be reconciled consistently from the released SEC evidence without unsupported imputation.

6.5 Available Phase-2 findings

The release establishes broad construction feasibility and one explicit non-estimability result. It does not persist complete factor IC, rank IC, standard error, t-statistic, and false-discovery evidence sufficient for H1. H1 therefore remains inconclusive.

7. Asset-Pricing Models and Results

7.1 CAPM

CAPM supplies the market-only benchmark (Sharpe, 1964). Its mean adjusted R-squared is 0.063 across 123 diagnostic portfolios.

7.2 Fama-French 3

The three-factor specification adds size and value (Fama and French, 1993).

7.3 Carhart 4

Momentum augments the three-factor model following Carhart (1997).

7.4 Fama-French 5

Profitability and investment extend the three-factor specification (Fama and French, 2015).

7.5 Configured q mapping

The project mapping uses log market capitalization, asset growth, and earnings yield as ME, IA, and ROE proxies. It is a study-specific implementation of q-motivated families, not a claim of exact replication of Hou, Xue, and Zhang (2015). No standalone custom model was estimated.

7.6 OLS and HAC inference

Coefficients are estimated by OLS. The reported intercept t-statistics use the released HAC/Newey-West covariance. The $p < .05$ column below is the share of the 123 diagnostic-portfolio intercept tests whose released HAC p-value is below 0.05. These in-sample intercepts are not interpreted as investable alpha. GRS and spanning statistics are unavailable in the released artifacts.

7.7 Empirical model comparison

Table 7. Asset-pricing model comparison and HAC intercept inference

Model	N	Mean adj. R2	Median adj. R2	Mean alpha	Median alpha	Mean HAC t	p<.05	Mean beta M
CAPM	123	0.063	0.061	0.01187	0.01128	2.467	85.4%	1.402
Fama-French 3	123	0.259	0.247	0.00923	0.00937	2.146	58.5%	1.421
Carhart 4	123	0.277	0.261	0.00899	0.00910	2.048	52.8%	1.408
Fama-French 5	123	0.301	0.281	0.01056	0.01026	2.569	92.7%	1.284
Configured q mapping	123	0.296	0.295	0.00875	0.00876	2.090	58.5%	1.291

Mean adjusted R-squared by model

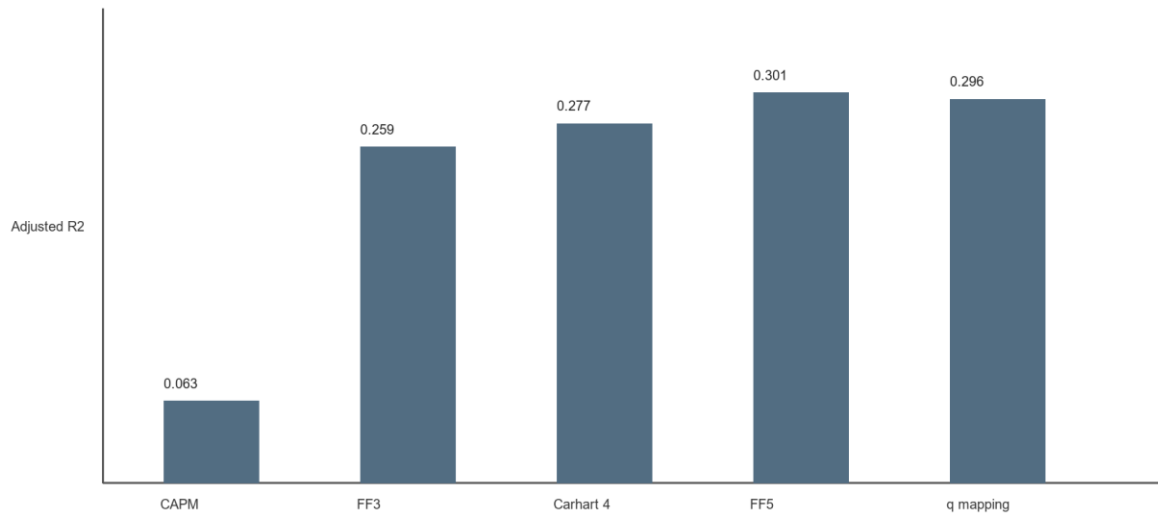


Figure 3. Mean adjusted R-squared by asset-pricing model.

FF5 exceeds CAPM by $0.301 - 0.063 = 0.238$ in mean adjusted R-squared. The comparison supports improved in-sample explanatory fit, not persistence or causal pricing success.

8. Portfolio Construction, Costs, and Risk

8.1 Eight methods

The comparison includes equal weight, minimum variance, mean variance, maximum Sharpe, maximum diversification, risk parity, hierarchical risk parity, and CVaR. No optimizer was rerun for this manuscript.

8.2 Constraints

All methods are long-only, fully invested, and unlevered, with monthly rebalancing. The tests use factor portfolios rather than direct investable securities.

8.3 Transaction costs

Table 8. Ex-ante one-way transaction-cost assumptions (basis points)

Component	LOW	BASE	HIGH
Commission	0	0	0
Spread	2	5	10
Slippage	1	2	4
Fixed impact allowance	2	3	6
Total	5	10	20

These are assumptions fixed before result inspection, not observed broker executions. Nonlinear market impact was not independently estimated.

8.4 Performance

Table 9. BASE portfolio performance, 2024-01-31 to 2026-08-04

Method	Annualized net return	Volatility	Sharpe	Maximum drawdown	Net cumulative
Equal weight	16.10%	11.14%	1.383	-8.57%	48.89%
Minimum variance	13.95%	10.00%	1.341	-6.69%	41.64%
Mean variance	12.72%	16.04%	0.818	-17.81%	37.60%
Maximum Sharpe	20.55%	12.30%	1.575	-7.51%	64.61%
Maximum diversification	17.31%	10.97%	1.499	-5.43%	53.08%
Risk parity	15.76%	10.79%	1.397	-8.33%	47.72%
Hierarchical risk parity	15.48%	10.56%	1.403	-8.13%	46.79%
CVaR	25.48%	14.21%	1.667	-8.55%	83.17%

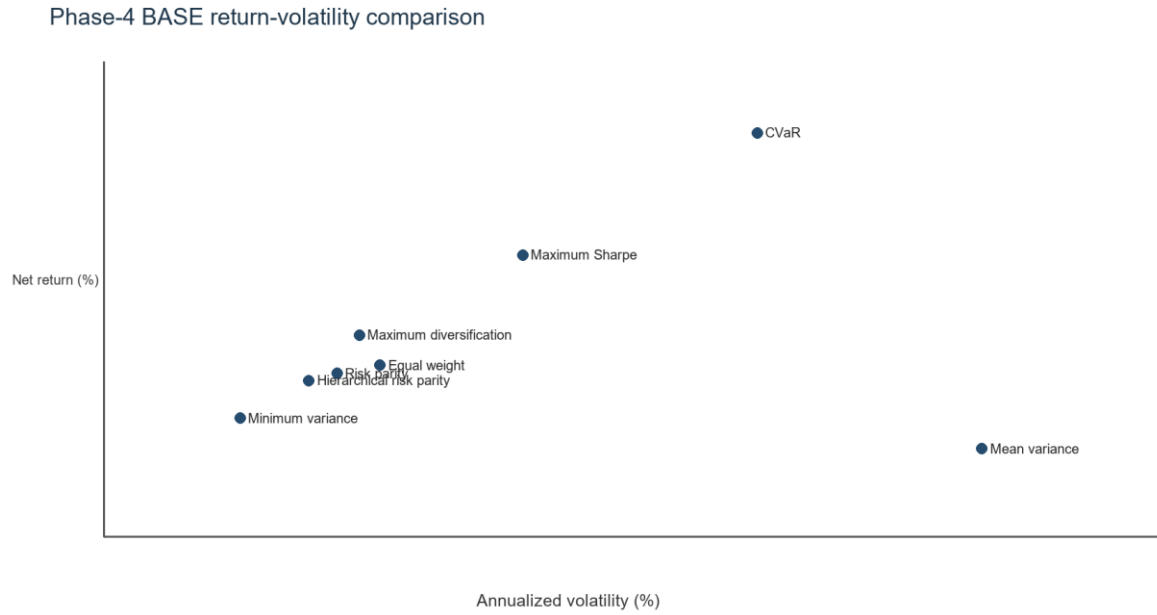


Figure 4. Phase-4 BASE return-volatility comparison.

8.5 Cost sensitivity

Table 10. Cumulative-return sensitivity to frozen costs

Method	Gross	LOW net	BASE net	HIGH net
Equal weight	49.12%	49.01%	48.89%	48.66%
Minimum variance	42.96%	42.30%	41.64%	40.34%
Mean variance	41.45%	39.52%	37.60%	33.85%
Maximum Sharpe	66.66%	65.63%	64.61%	62.59%
Maximum diversification	54.13%	53.60%	53.08%	52.03%
Risk parity	47.98%	47.85%	47.72%	47.47%
Hierarchical risk parity	47.42%	47.10%	46.79%	46.17%
CVaR	84.30%	83.73%	83.17%	82.06%

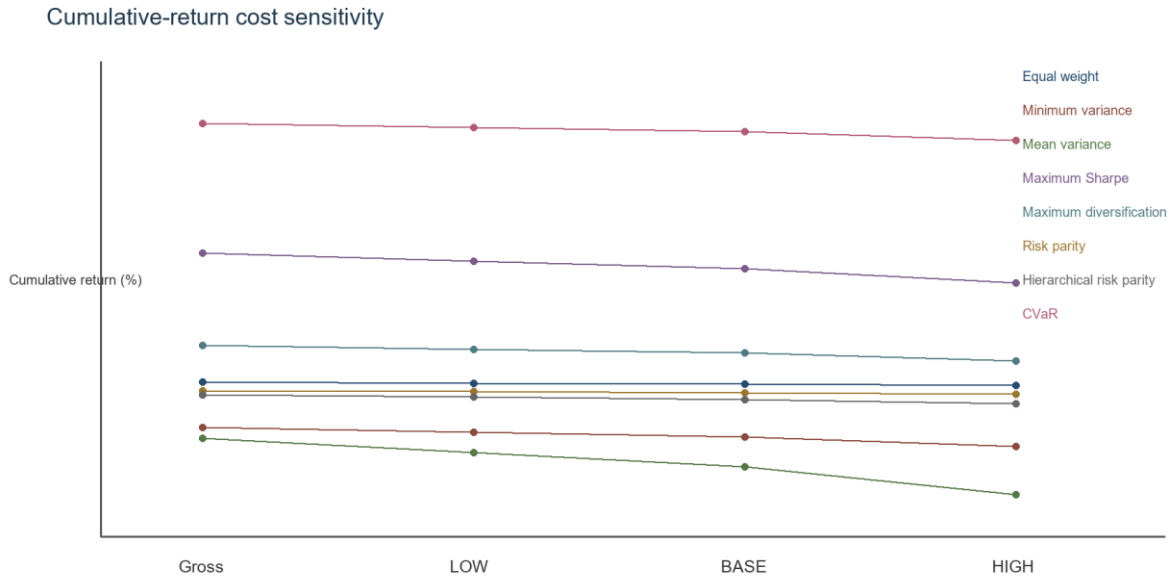


Figure 5. Cumulative-return cost sensitivity by method.

Higher costs reduce every cumulative result. Equal weight's BASE drag is 49.12% - 48.89% = 0.23 percentage points. Mean variance's HIGH drag is 41.45% - 33.85% = 7.60 percentage points.

8.6 Risk metrics

Table 11. Exact released BASE risk metrics

Method	Net cum.	Gross cum.	Vol.	Sharpe	Sortino	MDD	VaR95	CVaR95	Track err.	Info ratio	Beta
Equal Weight	48.89%	49.12%	11.14%	1.383	2.441	-8.57%	4.92%	5.38%	5.20%	-0.850	0.833
Minimum Variance	41.64%	42.96%	10.00%	1.341	2.136	-6.69%	4.55%	6.09%	10.03%	-0.641	0.498
Mean Variance	37.60%	41.45%	16.04%	0.818	1.153	-17.81%	4.53%	10.65%	13.55%	-0.495	0.753
Maximum Sharpe	64.61%	66.66%	12.30%	1.575	2.967	-7.51%	4.65%	5.85%	10.67%	-0.044	0.628
Maximum Diversification	53.08%	54.13%	10.97%	1.499	2.935	-5.43%	4.28%	4.94%	9.30%	-0.364	0.616
Risk Parity	47.72%	47.98%	10.79%	1.397	2.408	-8.33%	4.89%	5.39%	5.41%	-0.879	0.799
Hierarchical Risk Parity	46.79%	47.42%	10.56%	1.403	2.382	-8.13%	4.84%	5.43%	5.68%	-0.884	0.772
Cvar	83.17%	84.30%	14.21%	1.667	3.163	-8.55%	5.05%	7.58%	12.90%	0.298	0.622

8.7 Scenarios

Table 12. Released instantaneous scenario impacts

Scenario family	Impact
Market	-0.244%
Rates	-0.073%
Volatility	-0.122%

Scenario family	Impact
Inflation	-0.061%
Liquidity	-0.037%
Custom	+0.024%

Scenario impacts are exposure mappings, not forecasts.

9. Walk-Forward Machine Learning

9.1 Features and target

The final feature set contains book-to-market and 12-minus-1 momentum. The target is the security's 21-trading-session total return minus SPY's return over the same horizon.

9.2 Temporal split

The design has 60 training months, 12 validation months, and one test month, with monthly retraining. Purging removes overlapping label horizons and a one-month embargo separates validation from testing. The released decision panel contains 27,640 observations across 91 monthly dates; 17 complete out-of-sample folds cover 28 February 2025 through 30 June 2026.

9.3 Models

The eight rows are zero, historical mean, factor composite, linear, Ridge, Elastic Net, Random Forest, and XGBoost. Constant predictions make rank correlation undefined, so both constant baselines report IC as N/A rather than zero.

9.4 Tuning/search

Ridge alpha is selected from [0.1, 1.0, 10.0]. Elastic Net uses alpha [0.0001, 0.001] and l1_ratio [0.25, 0.5, 0.75]. Random Forest uses n_estimators [50, 100] and max_depth [3, 6]. XGBoost uses n_estimators [50, 100], max_depth [2, 3], and learning_rate [0.05, 0.1]. Validation Spearman IC is the selection criterion. Linear preprocessing is fit on training data only; seed 17 and one thread are fixed.

9.5 Bootstrap methodology

Mean fold IC is bootstrapped over monthly/fold IC observations with 1,000 replications, a 95% interval, and seed 17. The small number of complete folds is reflected in wide intervals.

9.6 Predictive results

Table 13. Exact eight-model predictive results

Model	Mean IC	95% bootstrap CI	Positive IC folds	Positive rate
Zero baseline	N/A	N/A	N/A	N/A
Historical mean	N/A	N/A	N/A	N/A

Model	Mean IC	95% bootstrap CI	Positive IC folds	Positive rate
Factor composite	0.017412	[-0.026401, 0.110608]	10/17	58.8%
Linear	0.017412	[-0.026401, 0.110608]	10/17	58.8%
Ridge	0.017414	[-0.026410, 0.110637]	10/17	58.8%
Elastic Net	0.027535	[-0.024701, 0.151649]	11/17	64.7%
Random Forest	0.009558	[-0.074074, 0.131357]	10/17	58.8%
XGBoost	-0.009069	[-0.049268, 0.055085]	8/17	47.1%

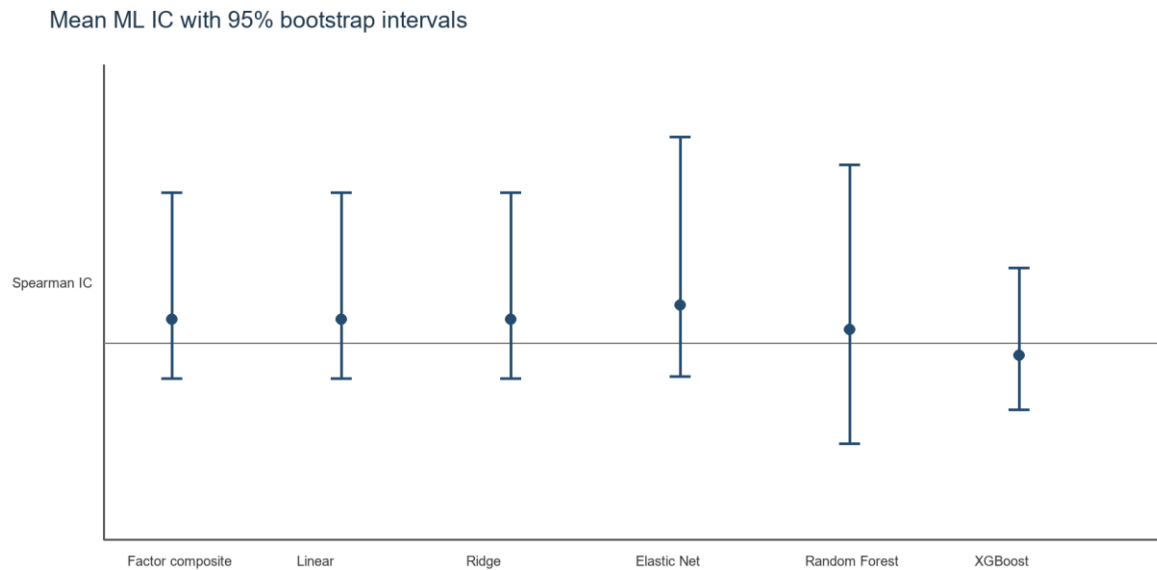


Figure 6. Mean ML information coefficients with 95% bootstrap intervals.

Every non-constant model interval includes zero. Stable predictive information is not established.

9.7 Economic results

Table 14. Exact eight-model after-cost economic fold results

Model	Mean net fold return	Median net fold return	Positive economic folds
Zero baseline	-0.001415	-0.001376	8/17
Historical mean	-0.001415	-0.001376	8/17
Factor composite	-0.000973	-0.000849	7/17
Linear	-0.000973	-0.000849	7/17
Ridge	-0.000973	-0.000849	7/17
Elastic Net	-0.001039	-0.000333	7/17
Random Forest	+0.000074	+0.000942	9/17
XGBoost	-0.000044	+0.001838	11/17

Random Forest has a small positive mean and 9 positive folds, while XGBoost has a positive median but a negative mean. Neither pattern establishes stable after-cost incremental value.

9.8 Explanations

Linear-family coefficients rank book-to-market above momentum. Random Forest permutation importance ranks book-to-market first while its SHAP summary ranks momentum first. XGBoost ranks book-to-market first under both methods. These are fitted-model attribution measures, not causal estimates; the inconsistency supports H7 as not supported.

10. Robustness and Research-Integrity Checks

The released design includes a common benchmark calendar, LOW/BASE/HIGH cost sensitivity, multiple portfolio methods, six scenario mappings, constant and model-based ML challengers, purged temporal folds, bootstrap intervals, feature ablations, drift diagnostics, and point-in-time joins. The released implementation also passed software-integrity, restart, and publication-authentication checks; technical details are reported in Appendix J.

11. Hypothesis Assessment

Table 15. Final hypothesis assessment

Hypothesis	Classification	Basis
H1	Inconclusive	Complete factor-level cost and multiplicity decision is not persisted.
H2	Partially supported	Multifactor models improve in-sample adjusted R-squared; persistent alpha is not established.
H3	Inconclusive	Rolling evidence exists, but the complete original temporal-stability decision is absent.
H4	Supported	Every higher cost setting reduces cumulative results.
H5	Partially supported	Methods alter observed risk-return profiles; durable superiority is not established.
H6	Not supported	All non-constant IC intervals include zero and economic gains are unstable.
H7	Not supported	Attribution rankings vary across methods and are noncausal.

12. Discussion

12.1 Factor and asset-pricing evidence

The selected public-data panel supports broad characteristic construction, but feasibility is not significance. Multifactor models explain more in-sample variation than CAPM on the project's diagnostic portfolios. The result is consistent with established factor structures, although the sample and test assets limit external validity.

12.2 Portfolio evidence

Allocation rules produce materially different observed volatility, drawdown, benchmark tracking, and tail risk. CVaR has the highest observed cumulative return in the short window, while minimum variance has the lowest volatility; neither is labelled the best strategy because persistence has not been established.

12.3 Transaction costs

The monotonic cost results are the most direct economic finding. They show that even simple pre-specified allowances change cumulative comparisons, especially for turnover-sensitive allocations. Because the costs are assumptions rather than executions, they measure sensitivity rather than realized implementation shortfall.

12.4 Why the negative ML result matters

The ML evidence resists a common temptation to tune until a positive narrative emerges. With only two features and 17 folds, estimates are imprecise; all IC intervals include zero. Preserving that outcome distinguishes an empirical test from a model-selection exercise optimized for presentation.

12.5 Implications for public-data research

Public sources can support useful, traceable empirical work when availability and source changes are explicit. They cannot substitute for absent upstream securities or standardized proprietary histories. The appropriate contribution is therefore source-bounded evidence, not equivalence to CRSP, Compustat, or historical-index databases.

13. Limitations

- The source-selected universe is not CRSP-, Compustat-, historical S&P 500-, or Russell-replication evidence.
- HF Data Library warns that its pre-2022 fixed universe omits provider-absent delisted firms; project controls cannot reconstruct them.
- The March-2022 PiTrading-to-IEX transition creates a volume and liquidity break; IEX represents approximately 2–3% of consolidated volume.
- Cross-sectional breadth is sparse in 2018–2019, and SEC concepts remain heterogeneous despite point-in-time availability.
- Equity issuance is not estimable, and unresolved identity, terminal-event, share, and splice candidates are excluded.
- Phase 4 covers only 32 months and uses factor-portfolio test assets. Costs are assumed rather than observed, and nonlinear market impact is not estimated.
- Phase 5 has two features and 17 complete folds; all aggregate IC intervals include zero. Explanation methods are noncausal.
- H1 and H3 remain incomplete under their original decision rules.

14. Future Research

Future work should use a point-in-time, delisting-inclusive security master; independently reconcile consolidated volume across the 2022 source boundary; extend the out-of-sample portfolio window; incorporate observed execution data where lawfully available; and add pre-specified characteristics only after the present evidence is frozen. A larger fold count would permit more informative ML uncertainty estimates. None of these extensions is inferred from the current results.

15. Conclusion

A connected public-data pipeline can support broad factor construction, robust asset-pricing comparison, constrained portfolio analysis, and temporally ordered prediction while preserving uncertainty. In this sample, multifactor models improve in-sample explanatory fit, portfolio methods produce different observed risk profiles, and higher assumed costs reduce every cumulative result. The analysis does not establish stable factor significance under H1, temporal stability under H3, persistent portfolio superiority, or stable machine-learning value. Those boundaries are part of the result.

Data and Code Availability

Public repository: <https://github.com/amitdudi04/institutional-multi-factor-asset-pricing-platform>

Empirical tag: project-complete-public-data-v1

Empirical commit: 5a3e930665756fa7aaedeceb5f9ab90792bcf849

Code and configuration are public. Third-party data are not redistributed, and exact reproduction requires lawful access to the source evidence. The repository's MIT licence does not supersede source licences. Publication identifiers and checksums support artifact-level verification.

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Competing Interests

The author declares no competing interests.

Declaration of Generative AI and AI-Assisted Technologies

The author used AI-assisted tools under direct human supervision for software-development support, debugging, document structuring, and language refinement during preparation of the

project and manuscript. The author defined and approved the research questions, data-source choices, empirical specifications, validation rules, and final interpretations; reviewed the implementation and generated outputs; and verified the numerical results reported in this manuscript against the project's versioned empirical artifacts. The author takes full responsibility for the analysis, interpretation, conclusions, and final text. No AI system is listed as an author.

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Appendix A. Complete 48-Factor Definition Catalog

Factor ID	Observations	Start	End	Estimability
accruals	715,447	2018-02-14	2026-08-04	ESTIMABLE
amihud_illiquidity	715,447	2018-02-14	2026-08-04	ESTIMABLE
asset_growth	715,447	2018-02-14	2026-08-04	ESTIMABLE
asset_turnover	715,447	2018-02-14	2026-08-04	ESTIMABLE
average_dollar_volume	715,447	2018-02-14	2026-08-04	ESTIMABLE
beta	715,447	2018-02-14	2026-08-04	ESTIMABLE
bid_ask_proxy	715,447	2018-02-14	2026-08-04	ESTIMABLE
book_to_market	715,447	2018-02-14	2026-08-04	ESTIMABLE
capex_growth	715,447	2018-02-14	2026-08-04	ESTIMABLE
cash_flow_yield	715,447	2018-02-14	2026-08-04	ESTIMABLE
debt_to_equity	715,447	2018-02-14	2026-08-04	ESTIMABLE
dividend_yield	715,447	2018-02-14	2026-08-04	ESTIMABLE
downside_beta	715,447	2018-02-14	2026-08-04	ESTIMABLE
downside_volatility	715,447	2018-02-14	2026-08-04	ESTIMABLE
earnings_yield	715,447	2018-02-14	2026-08-04	ESTIMABLE
equity_issuance	715,447	2018-02-14	2026-08-04	NOT ESTIMABLE
excess_return	715,447	2018-02-14	2026-08-04	ESTIMABLE
gross_profitability	715,447	2018-02-14	2026-08-04	ESTIMABLE
idiosyncratic_volatility	715,447	2018-02-14	2026-08-04	ESTIMABLE
interest_coverage	715,447	2018-02-14	2026-08-04	ESTIMABLE
leverage	715,447	2018-02-14	2026-08-04	ESTIMABLE
log_market_cap	715,447	2018-02-14	2026-08-04	ESTIMABLE
market_cap	715,447	2018-02-14	2026-08-04	ESTIMABLE
market_return	715,447	2018-02-14	2026-08-04	ESTIMABLE
momentum_12_1m	715,447	2018-02-14	2026-08-04	ESTIMABLE
momentum_12m	715,447	2018-02-14	2026-08-04	ESTIMABLE
momentum_1m	715,447	2018-02-14	2026-08-04	ESTIMABLE
momentum_24m	715,447	2018-02-14	2026-08-04	ESTIMABLE
momentum_3m	715,447	2018-02-14	2026-08-04	ESTIMABLE
momentum_6m	715,447	2018-02-14	2026-08-04	ESTIMABLE
net_margin	715,447	2018-02-14	2026-08-04	ESTIMABLE
operating_margin	715,447	2018-02-14	2026-08-04	ESTIMABLE
price_to_book	715,447	2018-02-14	2026-08-04	ESTIMABLE
residual_momentum	715,447	2018-02-14	2026-08-04	ESTIMABLE
risk_free_rate	715,447	2018-02-14	2026-08-04	ESTIMABLE
roa	715,447	2018-02-14	2026-08-04	ESTIMABLE
roe	715,447	2018-02-14	2026-08-04	ESTIMABLE
rolling_beta	715,447	2018-02-14	2026-08-04	ESTIMABLE
rolling_correlation	715,447	2018-02-14	2026-08-04	ESTIMABLE
rolling_market_beta	715,447	2018-02-14	2026-08-04	ESTIMABLE
rolling_volatility	715,447	2018-02-14	2026-08-04	ESTIMABLE

Factor ID	Observations	Start	End	Estimability
semi_variance	715,447	2018-02-14	2026-08-04	ESTIMABLE
size_large	715,447	2018-02-14	2026-08-04	ESTIMABLE
size_mid	715,447	2018-02-14	2026-08-04	ESTIMABLE
size_small	715,447	2018-02-14	2026-08-04	ESTIMABLE
tracking_error	715,447	2018-02-14	2026-08-04	ESTIMABLE
turnover	715,447	2018-02-14	2026-08-04	ESTIMABLE
working_capital_growth	715,447	2018-02-14	2026-08-04	ESTIMABLE

Appendix B. Detailed Universe Screening and Annual Breadth

Screening retained 487 and rejected 335 candidates after attribution, splice, identity, share-basis, terminal-event, asset-type, venue, and fundamental-evidence checks. Annual breadth is reproduced in Table 4; the complete daily evidence remains in the frozen market publication.

Stage	Start	End	Frequency
Phase 1 market	2018-02-14	2026-08-04	Daily
Phase 1 fundamentals (available_at)	2019-02-23	2026-08-15	Event time
Phase 2 factors	2018-02-14	2026-08-04	Daily/monthly formation
Phase 3 asset pricing	2018-03-29	2026-08-04	Monthly
Phase 4 portfolios	2024-01-31	2026-08-04	Monthly
Phase 5 decisions	2025-02-28	2026-06-30	Monthly

Appendix C. Additional Descriptive Statistics

Factor ID	N	Mean	SD	P1	P25	Median	P75	P99
book_to_market	585,078	0.395473	0.440632	-0.201907	0.118791	0.292552	0.557899	1.982315
momentum_12_1m	593,210	0.124992	0.444258	-0.601507	-0.118453	0.069121	0.283342	1.645525
rolling_volatility	706,194	0.389935	0.209722	0.157417	0.252073	0.329544	0.462254	1.167571

Appendix D. Asset-Pricing Model Specifications and Extended Inference

Table 7 reports all persisted cross-model summary fields. Classical, HC0–HC3, and HAC covariance outputs and rolling coefficients remain in the released Phase-3 artifacts. GRS and spanning statistics are not available.

Appendix E. Complete Portfolio and Risk Results

Tables 9–12 report the complete uniformly persisted cross-method performance, cost, risk, and scenario evidence. HHI, turnover, and risk-contribution summaries are omitted from the cross-method table because they are not uniformly persisted.

Appendix F. Transaction-Cost and Worked Calculations

Calculation	Result
Acceptance	$487 / 822 = 59.25\%$
Rejection	$335 / 822 = 40.75\%$
FF5 fit improvement	$0.301 - 0.063 = 0.238$
Equal-weight BASE drag	$49.12\% - 48.89\% = 0.23$ percentage points
Mean-variance HIGH drag	$41.45\% - 33.85\% = 7.60$ percentage points

Appendix G. Complete Eight-Model ML Results and Hyperparameter Grid

Tables 13–14 retain one row per model. Frozen grids are Ridge alpha [0.1, 1.0, 10.0]; Elastic Net alpha [0.0001, 0.001] and l1_ratio [0.25, 0.5, 0.75]; Random Forest n_estimators [50, 100] and max_depth [3, 6]; XGBoost n_estimators [50, 100], max_depth [2, 3], learning_rate [0.05, 0.1].

Appendix H. Reproducibility Identifiers and Publication Lineage

Stage	Publication ID
Phase 1 market	factor_market_input-89848f8e3c34d53d0432f6e0
Phase 1 fundamentals	factor_fundamental_input-962c8ada3e74e32b00568cd7
Phase 2	factor-5a73616f131fca2102fa2ca4141ad64f
Phase 3	asset-pricing-c0a72fae5349f1048e18557250da1927
Phase 4 BASE parent	portfolio-dfef8d60d8a771e61224f0b314efd86c
Phase 4 scenarios	portfolio-2bbcdf643c1ca3e00f579e0011180fa

The eight definitive ML publication IDs and four report IDs are enumerated in the versioned reproducibility guide. The empirical tag and commit appear in Data and Code Availability.

Appendix I. Detailed H1–H7 Evidence Matrix

Hypothesis	Required evidence	Available evidence	Classification
H1	At least one factor spread survives the complete pre-specified cost and multiplicity rule	Complete factor-level cost and multiplicity decision is not persisted.	Inconclusive
H2	Multifactor models improve explanatory fit relative to CAPM	Multifactor models improve in-sample adjusted R-squared; persistent alpha is	Partially supported

Hypothesis	Required evidence	Available evidence	Classification
		not established.	
H3	Estimated relations are stable through time under the original rule	Rolling evidence exists, but the complete original temporal-stability decision is absent.	Inconclusive
H4	Higher transaction-cost assumptions reduce cumulative outcomes	Every higher cost setting reduces cumulative results.	Supported
H5	Constrained methods alter risk-return trade-offs	Methods alter observed risk-return profiles; durable superiority is not established.	Partially supported
H6	ML adds stable OOS predictive and after-cost value	All non-constant IC intervals include zero and economic gains are unstable.	Not supported
H7	Explanation rankings are stable across methods	Attribution rankings vary across methods and are noncausal.	Not supported

Appendix J. Software Verification

The verified software release records 301 passing tests and 90.52% branch-aware coverage. Lint, formatting, strict type checks, dependency-lock verification, catalog validation, restart checks, and publication-integrity checks passed. These controls support software correctness and reproducibility; they do not constitute evidence of investment performance or persistent alpha. Docker and repository-hosting constraints are operational matters, not empirical limitations.

Appendix K. Dashboard Interface Note

The Streamlit dashboard is a presentation layer over stored research publications. Dashboard screenshots are intentionally omitted from this manuscript because interface labels can change independently from the underlying empirical results. The current interface scope, startup commands, and delivery boundaries are documented in the repository's docs/SOFTWARE_USAGE.md.

Appendix K. Dashboard Interface Note (continued)

Dashboard scope

The current dashboard organizes the research outputs into the following views: Platform Overview; Data and Lineage Explorer; Factor Research; Asset-Pricing Research; Portfolio Construction; Risk Analytics; Machine Learning; Validation; and Report Builder. These views read stored research results and provenance metadata rather than re-computing finance logic inside the presentation layer.

Review boundary

The dashboard is not a brokerage, order-routing, or live-trading system. A missing publication is shown as unavailable rather than replaced with a synthetic result. For empirical review, the manuscript tables and docs/RESULTS.md remain the primary numerical record.

Appendix K. Dashboard Interface Note (continued)

Interpretation

Interface presentation should not be treated as additional statistical evidence. The dashboard summarizes the same stored factor, asset-pricing, portfolio, risk, and machine-learning publications described elsewhere in the paper. Model explanations are descriptive diagnostics and are not interpreted causally.

Reproducibility

The dashboard consumes the same versioned research publications and configuration identities used by the command-line and API layers. Exact empirical reproduction still requires lawful access to the third-party source data and mappings described in the repository documentation.

Appendix K. Dashboard Interface Note (continued)

Repository documentation

Current dashboard usage is documented in docs/SOFTWARE_USAGE.md, while architecture and research-result interpretation are documented in docs/ARCHITECTURE.md and docs/RESULTS.md. This separation keeps interface changes from being mistaken for changes to the empirical study.

Appendix L. Formula Reference

Equations (1)–(26) define the empirical transformations used in the study. Their surrounding sections cite the established methods. The benchmark-calendar reconciliation error in the released factor publication is 8.33×10^{-17} , consistent with numerical roundoff.

Appendix M. Numerical Provenance Statement

The final evidence ledger contains 249 controlled records: the prior 225 records covering empirical values, model settings, provenance, and explicitly unavailable evidence, plus 24 corrected long-form descriptive-statistic records for book-to-market, 12-minus-1 momentum, and rolling volatility. The final numerical audit classifies manuscript numbers as direct, derived, methodological constants, or reference values.

$$Rank(i, t) = rank[X(i, t)] / N(t) \quad (1)$$

$$F(t) = R(high, t) - R(low, t) \quad (2)$$

$$R(p, t) - R(f, t) = alpha(p) + beta(M, p)[R(M, t) - R(f, t)] + epsilon(p, t) \quad (3)$$

$$R(p, t) - R(f, t) = alpha + beta(M)MKT + beta(S)SMB + beta(H)HML + epsilon \quad (4)$$

$$Carhart4 = FF3 + beta(MOM)MOM \quad (5)$$

$$FF5 = alpha + beta(M)MKT + beta(S)SMB + beta(H)HML + beta(R)RMW + beta(C)CMA + epsilon \quad (6)$$

$$q = \alpha + \beta(M)MKT + \beta(ME)ME + \beta(IA)IA + \beta(ROE)ROE + \epsilon \quad (7)$$

$$\hat{\beta} = (X'X)^{-1}X'y \quad (8)$$

$$\text{Adjusted } R^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1) \quad (9)$$

$$\min(w) w' \Sigma w \quad (10)$$

$$\max(w) w' \mu - (\lambda/2) w' \Sigma w \quad (11)$$

$$\max(w) [w' \mu - R(f)] / \sqrt{w' \Sigma w} \quad (12)$$

$$DR(w) = w' \sigma / \sqrt{w' \Sigma w} \quad (13)$$

$$RC(i) = w(i)(\Sigma w)(i) / \sqrt{w' \Sigma w} \quad (14)$$

$$\text{Turnover}(t) = \text{one half } \sum(i) |w(\text{target}, i, t) - w(\text{pretrade}, i, t)| \quad (15)$$

$$\text{Cost}(t) = \text{Turnover}(t) \text{ times } c(t) \quad (16)$$

$$R(\text{net}, t) = R(\text{gross}, t) - \text{Cost}(t) \quad (17)$$

$$\text{VaR}(\alpha, L) = \inf\{\ell : P(L \leq \ell) \geq \alpha\} \quad (18)$$

$$\text{CVaR}(\alpha, L) = E[L | L \geq \text{VaR}(\alpha, L)] \quad (19)$$

$$\sigma(\text{annual}) = \sigma(\text{period})\sqrt{K} \quad (20)$$

$$\text{Sharpe} = E[R(p) - R(f)] / \sigma[R(p) - R(f)] \text{ times } \sqrt{K} \quad (21)$$

$$MDD = \min(t) \{V(t) / \max(s \leq t) V(s) - 1\} \quad (22)$$

$$Y(i, t) = R(i, t \text{ to } t + 21) - R(\text{SPY}, t \text{ to } t + 21) \quad (23)$$

$$\text{Ridge: } \min(\beta) \sum(i) (y(i) - x(i)' \beta)^2 + \lambda \|\beta\|_2^2 \quad (24)$$

$$\text{Elastic Net: } \min(\beta) \sum(i) (y(i) - x(i)' \beta)^2 + \lambda [a \|\beta\|_1 + (1 - a) \|\beta\|_2^2] \quad (25)$$

$$IC(t) = \text{Spearman } \rho(\text{prediction}(i, t), \text{target}(i, t)) \quad (26)$$